Room: _____

Column: _____

Row: _____

Name and surname: _____

Student ID number: _____

Image-based Biometrics

written exam

2022-02-09

Total points: **100**
Time limit: **60 minutes**
There are 10 questions in this exam. Support your answers with arguments!
Please write legibly and grammatically correct! You can answer in English or Slovene.
Literature is not allowed!
Communication and use of any type of electronic device is forbidden!

1. Describe the main differences between the three levels of features for fingerprint recognition. (10)
2. Sketch the design of an optical FTIR-based sensor and explain on it how the image is generated. (10)
3. What conclusions did we obtain from iris recognition system when we experimented with images of the same person but with different pupil dilations (in terms of authentic and impostor distributions)? Sketch and explain. (10)
4. Sketch and explain HOG image descriptors. Where did we use them? What is the difference between sparse and dense arrangements? (10)
5. We said that the amount of information available for fusion gets reduced as one progresses along the various processing modules of a biometric system. Select one modality and sketch and explain this statement on it. (10)
6. How can we detect the presence of printed contact lenses in ocular images? Can the printing process affect the detection – and if yes, how? (10)
7. What is the main difference between Eigenfaces and Fisherfaces? Write the algorithm (in pseudo-code) that gives Eigenfaces as an output. (10)
8. Where did we use Borda Count Method and Logistic Regression Method? How do we get the first one from the second one? (10)
9. Name and explain the properties of curves giving FNMR and FMR dependence. Additionally, let's say that we are observing two curves that are intersecting at one point: how can we argument which one (which system) is better? (10)
10. Sketch and describe the method for extraction of singularity points. (10)